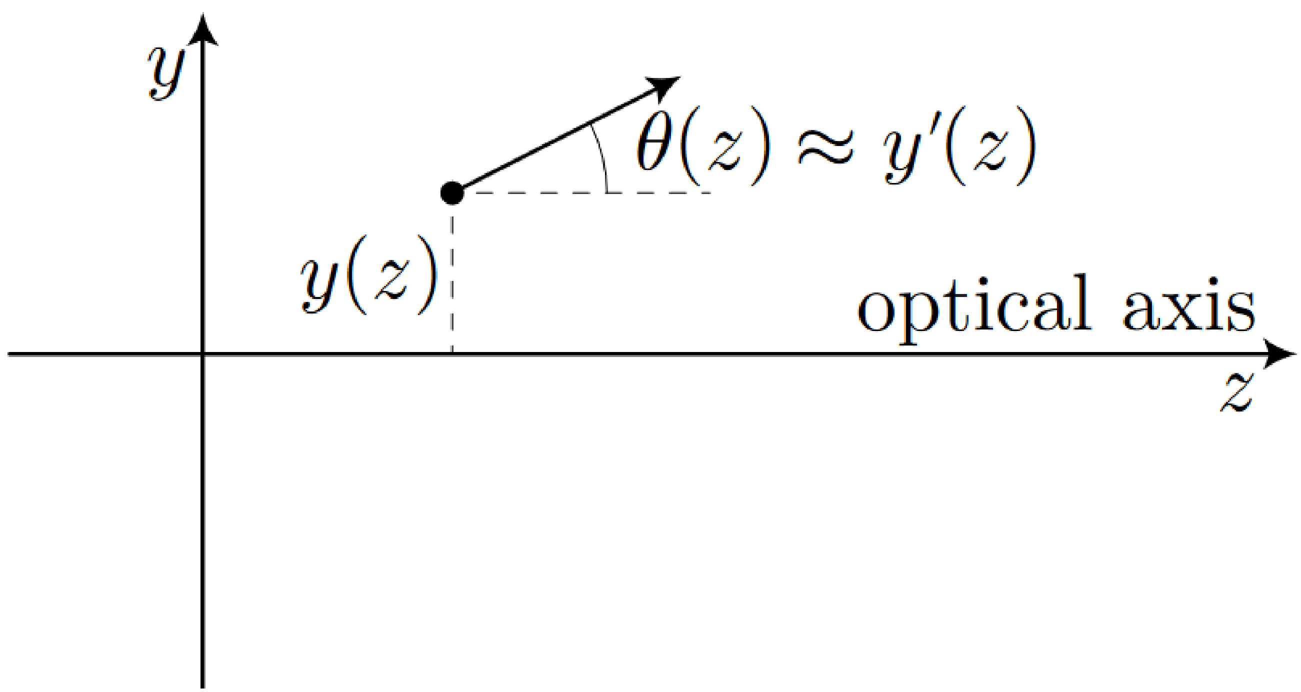
Ph134, Lecture 13

- Ray Optics:
  - Paraxial rays
  - Propagation of rays: ABCD matrices
  - Lensmaker's equation

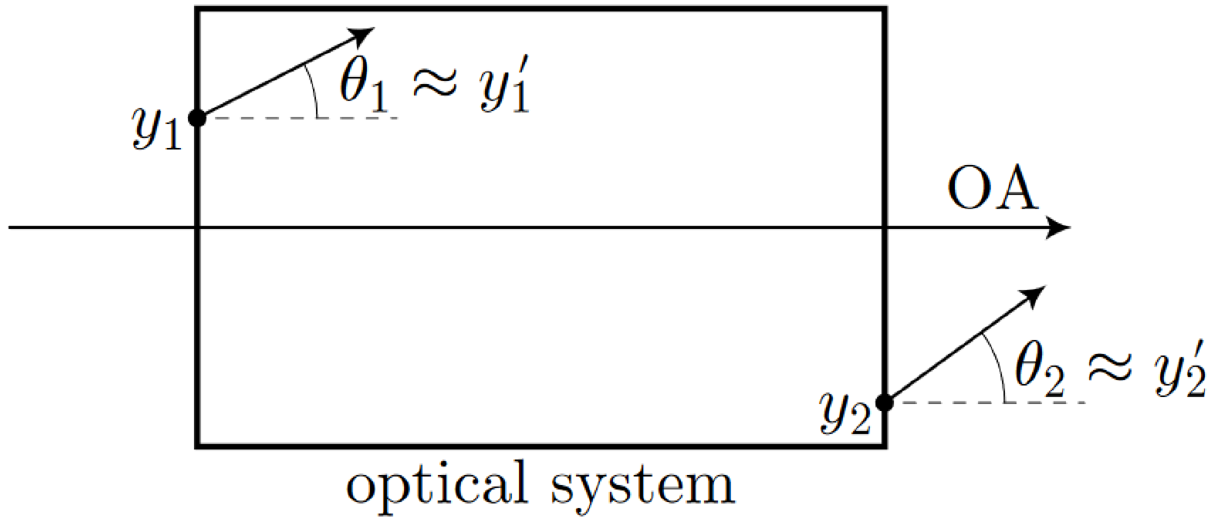
Paraxial Ray Optics

Wavefronts *are surfaces of constant phase.*
Rays *point in direction of propagation. This is the direction of fastest-increasing phase, so rays point perpendicular to* ***wavefronts****.*

For a beam:
Rays have y ≪ (propagation distances in z direction)
Their angle θ from the z axis is small.



Paraxial Ray Propagation

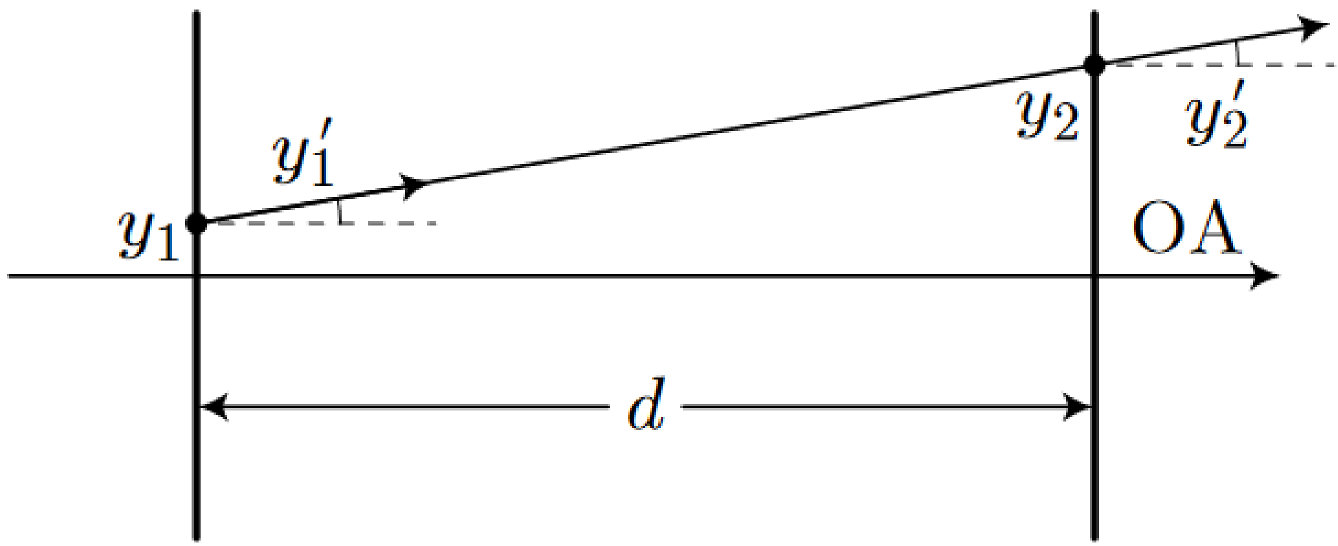


$$\begin{bmatrix} y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} f(y_1, \theta_1) \\ g(y_1, \theta_1) \end{bmatrix} \approx \begin{bmatrix} \left.\dfrac{\partial f}{\partial y_1}\right|_{0,0} y_1 + \left.\dfrac{\partial f}{\partial \theta_1}\right|_{0,0} \theta_1 \\ \left.\dfrac{\partial g}{\partial y_1}\right|_{0,0} y_1 + \left.\dfrac{\partial g}{\partial \theta_1}\right|_{0,0} \theta_1 \end{bmatrix} = \boxed{\begin{bmatrix} \left.\dfrac{\partial f}{\partial y_1}\right|_{0,0} & \left.\dfrac{\partial f}{\partial \theta_1}\right|_{0,0} \\ \left.\dfrac{\partial g}{\partial y_1}\right|_{0,0} & \left.\dfrac{\partial g}{\partial \theta_1}\right|_{0,0} \end{bmatrix}} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

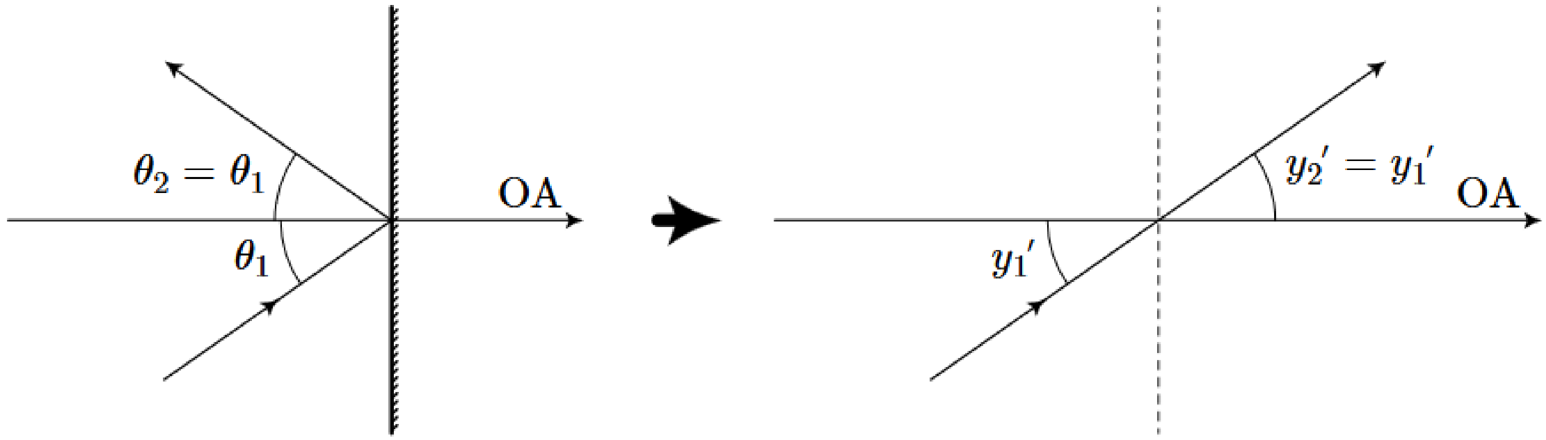
(the ABCD matrix for the optical system)

ABCD matrix for free-space propagation:



$$\mathbf{M} = \begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix}$$

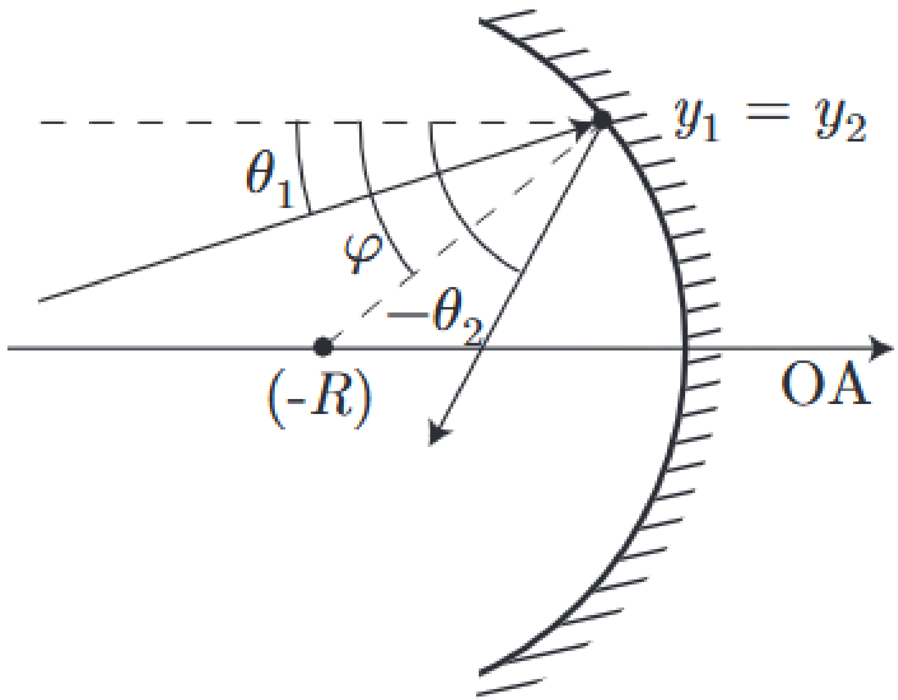
ABCD matrix for planar mirror:



"unfold" overall beam path so we effectively ignore the overall reversal of the beam's propagation direction

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

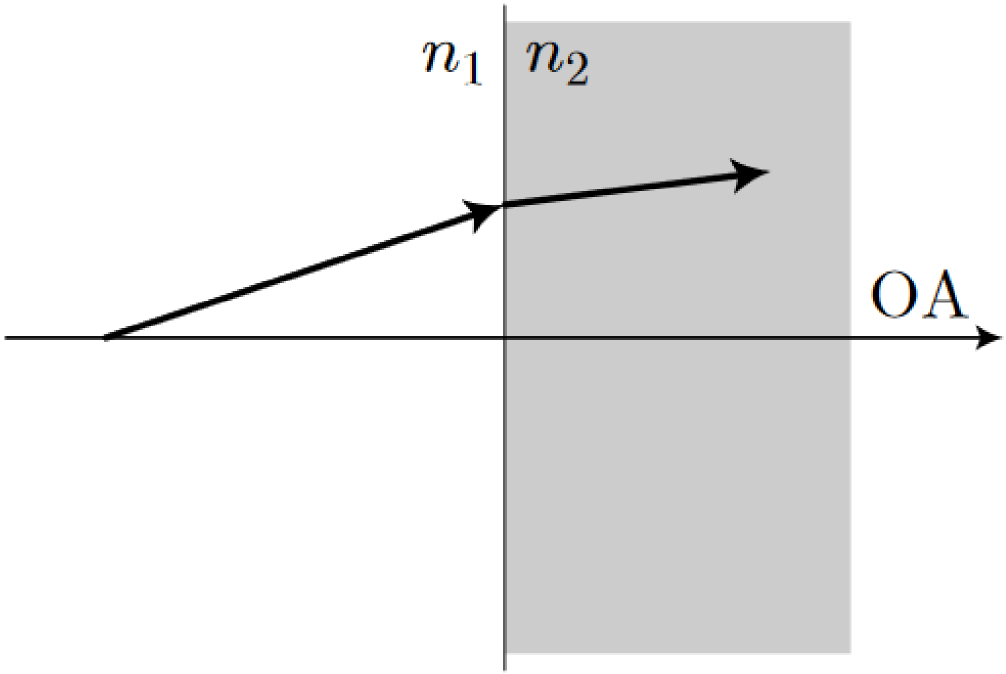
ABCD matrix for spherical mirror:



concave (shown): $R < 0$
convex: $R > 0$

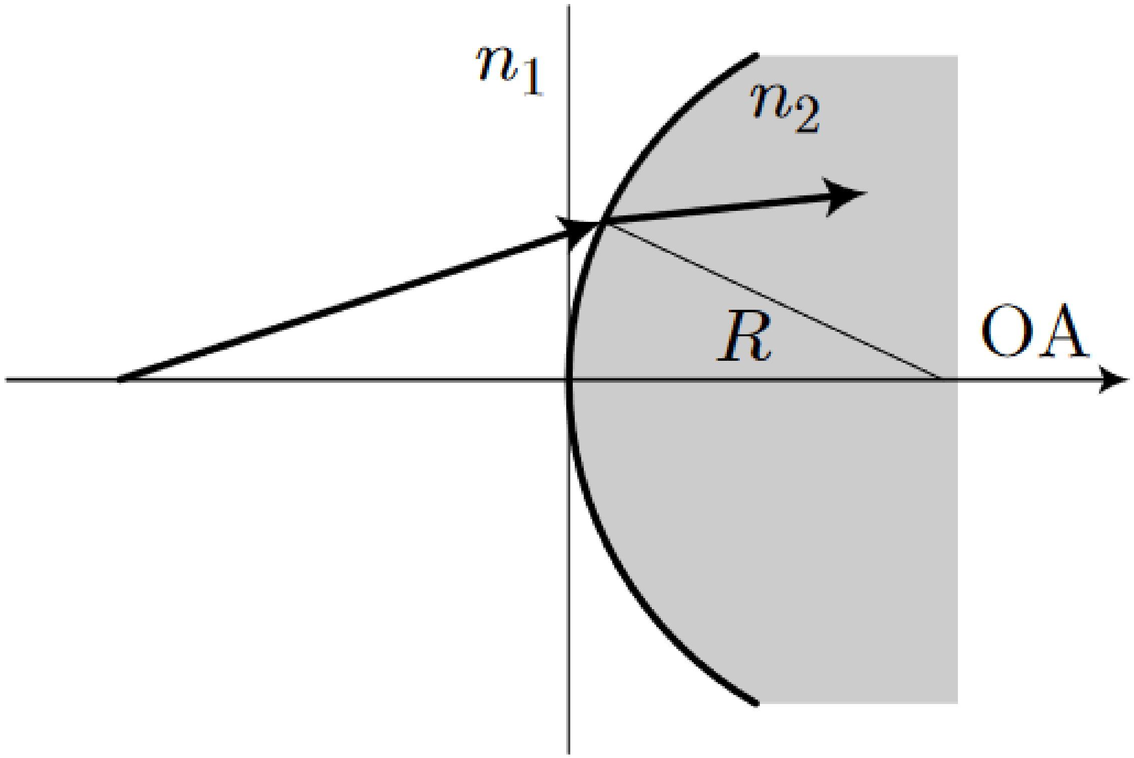
$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ \frac{2}{R} & 1 \end{bmatrix}$$

ABCD matrix for refractive interface ⊥ beam:



$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & n_1/n_2 \end{bmatrix}$$

ABCD matrix for spherical refractive interface:

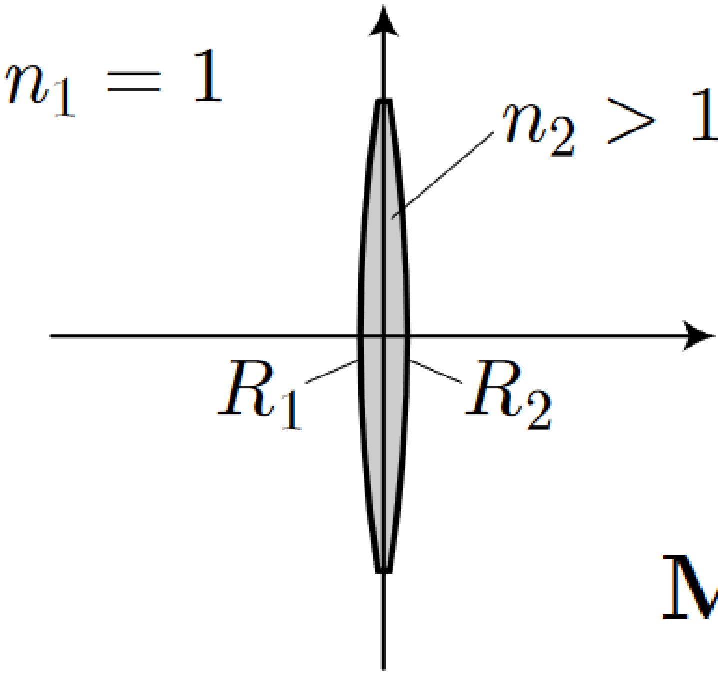


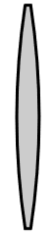
$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ -\dfrac{(n_2 - n_1)}{n_2 R} & \dfrac{n_1}{n_2} \end{bmatrix}$$

convex (shown): $R > 0$

concave: $R < 0$

ABCD matrix for thin lens:





$R_1 > 0, R_2 < 0$

$R_1 < 0, R_2 > 0$

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ \dfrac{(n_2 - n_1)}{n_1}\left(\dfrac{1}{R_2} - \dfrac{1}{R_1}\right) & 1 \end{bmatrix}$$

Call this quantity -1/f, where f is the ***focal length*** *of the lens.*

$$\frac{1}{f} = \frac{(n_{lens} - n_{medium})}{n_{medium}}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

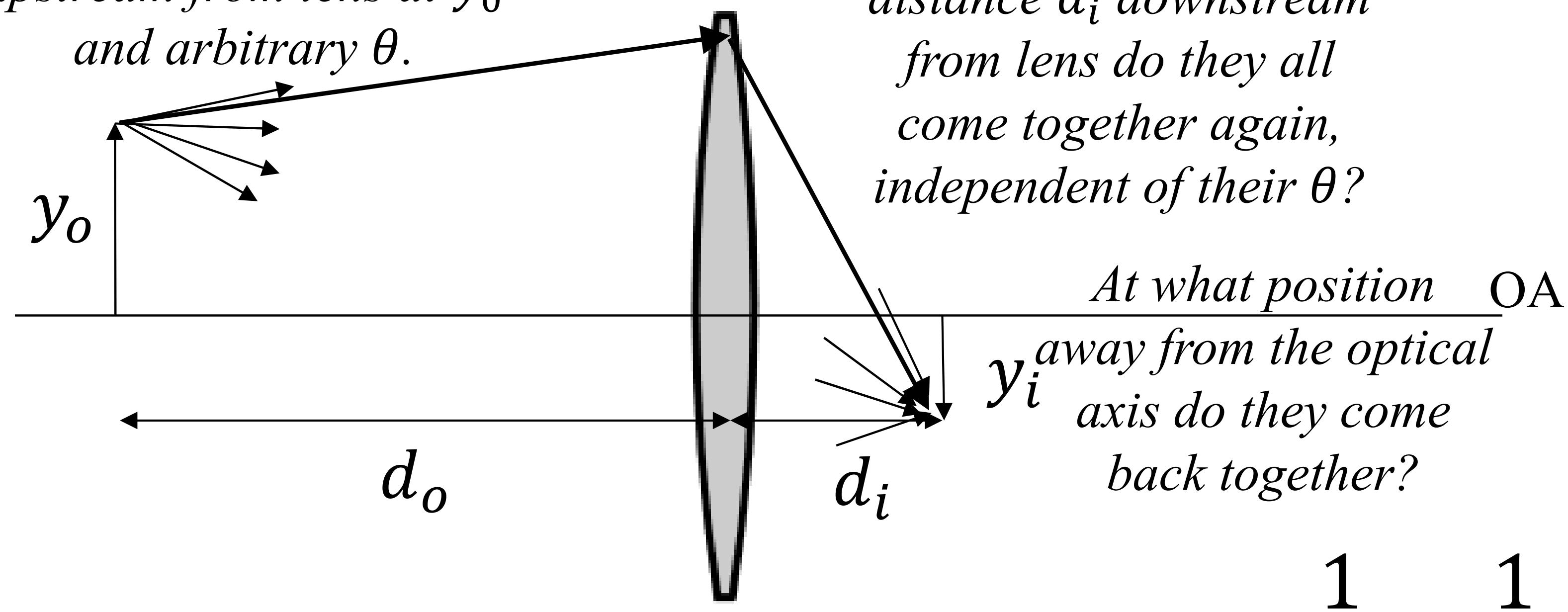
the lensmaker's equation

Thin lens equation and magnification:

Consider rays that start longitudinal distance d_0 upstream from lens at y_0 and arbitrary θ.

At what longitudinal distance d_i downstream from lens do they all come together again, independent of their θ?

At what position away from the optical axis do they come back together?



$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$\mathit{Magnification} \quad \frac{y_i}{y_o} = -\frac{d_i}{d_o}$$

Next time: what does f of a lens <u>mean</u>? Imaging with lenses / systems of lenses